

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

2bi2__ / 2IPbi2__

REGISTRATION
NUMBER

BIOLOGY

9648/02

Paper 2

26 August 2015

2 hours

Additional Materials: Answer Paper and Cover Page

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you submit.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

The number of marks is given in brackets [] at the end of each question or part question.

Sections A to C

Answer all questions.

Write your answers in this booklet.

Section D

Write your answers on the separate Answer Paper.

Begin each part of the question on a new piece of paper.

For Examiner's Use	
Sections A - C	[Total: 80]
1	10
2	10
3	10
4	9
5	11
6	11
7	9
8	10
Section D	[Total: 20]
9 / 10	20
Total	/ 100

This document consists of **20** printed pages, including this cover page and one blank page.

Section A

Answer **all** the questions in this section.

- 1** In beer-making, enzymes present in malting barley hydrolyse starch into fermentable sugars.

- (a)** Distinguish between the structure and function of starch and cellulose.

[3]

- (b)** Fig 1.1 shows the graph below shows the production of fermentable sugars at three different temperatures over a period of 60 minutes. All other conditions were controlled.

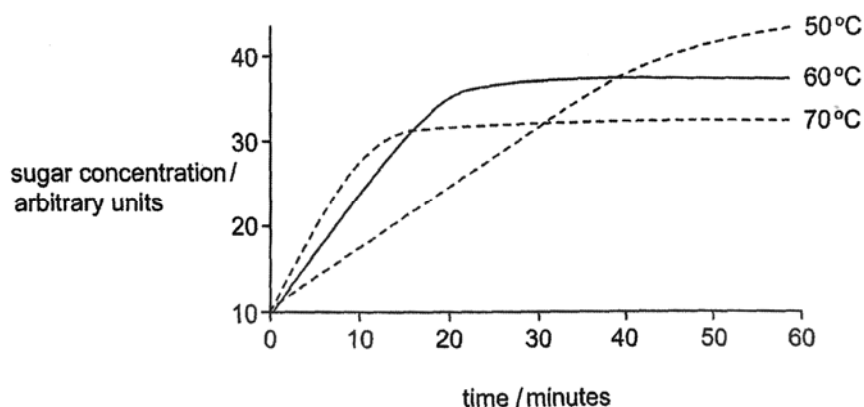


Fig. 1.1

- (i) State the enzyme used in the reaction above.

[1]

- (ii) Explain how the enzyme stated in (i) plays its role in the hydrolysis of starch into fermentable sugars.

[3]

- (iii) With reference to Fig. 1.1, explain the effect of increasing temperature on enzyme activity for the first 10 minutes of the reaction.

[3]

[Total: 10 m]

- 2 (a) DNA is replicated semi-conservatively.
State what is meant by semi-conservative replication.

[3]

In bacteria, it is common for two or more structural genes to be arranged together in an operon. Fig. 2.1 shows the structure of the *lac* operon and its control elements. Four different configuration of proteins binding to the control elements have been drawn for you.

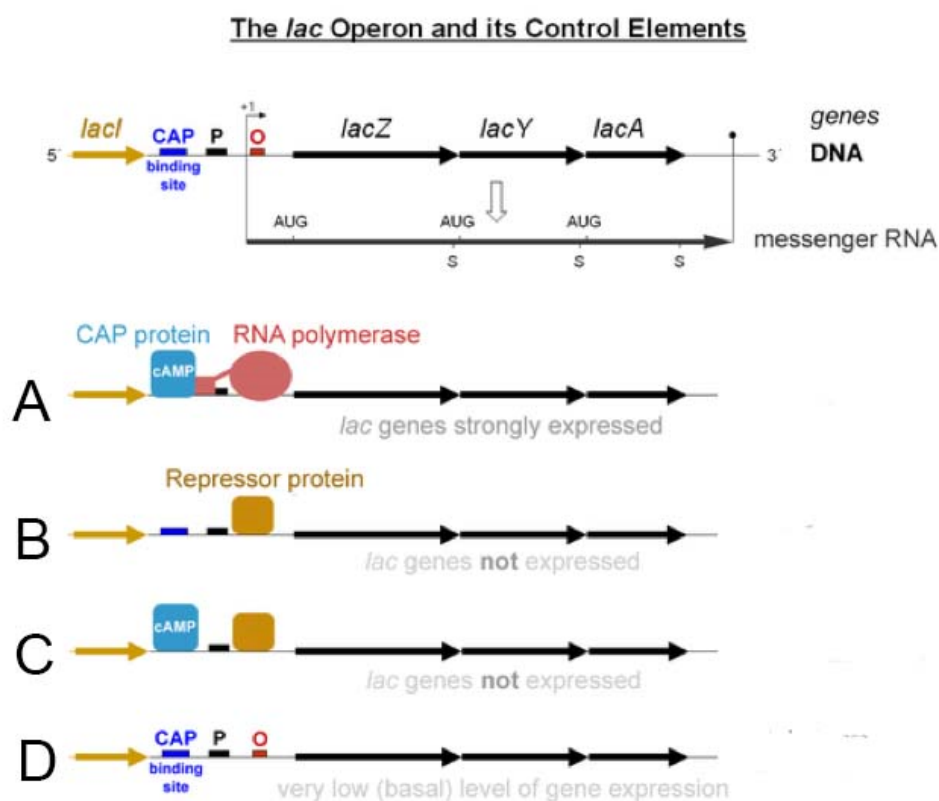


Fig. 2.1

- (b) Identify the four conditions (e.g. levels of glucose and lactose) that will result in configuration **A**, **B**, **C**, and **D**. (Please only use the keywords: high, low, unavailable, glucose, and lactose).

A:

B:

C:

D:

[4]

- (c) (i) A mutation that inactivates the *lac* repressor and prevents it from binding to the *lac* operator has been found. Explain the effect of this mutation on *lac* operon expression.

[2]

- (ii) Suggest a disadvantage to the bacterium for having this mutation.

[1]

[Total: 10 m]

- 3 (a) Control of gene expression in a eukaryote can take place at different levels. Briefly explain the control of gene expression at the three different level stated.

(i) Chromosomal level

[2]

(ii) Transcriptional level

[2]

(iii) Post-translational level

[2]

- (b) (i) In a laboratory, it was found that a particular cell line has a mutation in the *ras* gene promoter, leading to high expression of ras protein. However, cells from this cell line do not develop into tumour. Explain.

[2]

(ii) Other than the reason stated in (i), state two other causes that can lead to high level of expression of ras protein.

[2]

[Total: 10 m]

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Section B

Answer **all** the questions in this section.

- 4 A botanist during his investigation on the exotic *Mimosa pudica*, crossed a pure-bred plant with purple flowers and round leaves with a pure-bred plant with white flowers and oval leaves. All the F₁ generation had purple flowers and oval leaves.

A test cross was then made between the F₁ plant and a pure-bred plant with white flowers and round leaves. The phenotypes of 250 offspring of the test cross were recorded and shown below.

purple flowers, oval leaves : 21
 purple flowers, round leaves : 102
 white flowers, oval leaves : 110
 white flowers, round leaves : 17

- (a) Using A/a symbols to represent locus for flower colour and B/b symbols to represent locus for leaf shape, draw a genetic diagram to explain the results of the test cross.

[4]

(b) Explain why the expected dihybrid test cross ratio of 1:1:1:1 was not observed.

[2]

- (c) In another investigation on the height of the *Mimosa pudica*, the botanist crossed two plants with different stem heights. The results of the cross are shown in Fig. 4.1.

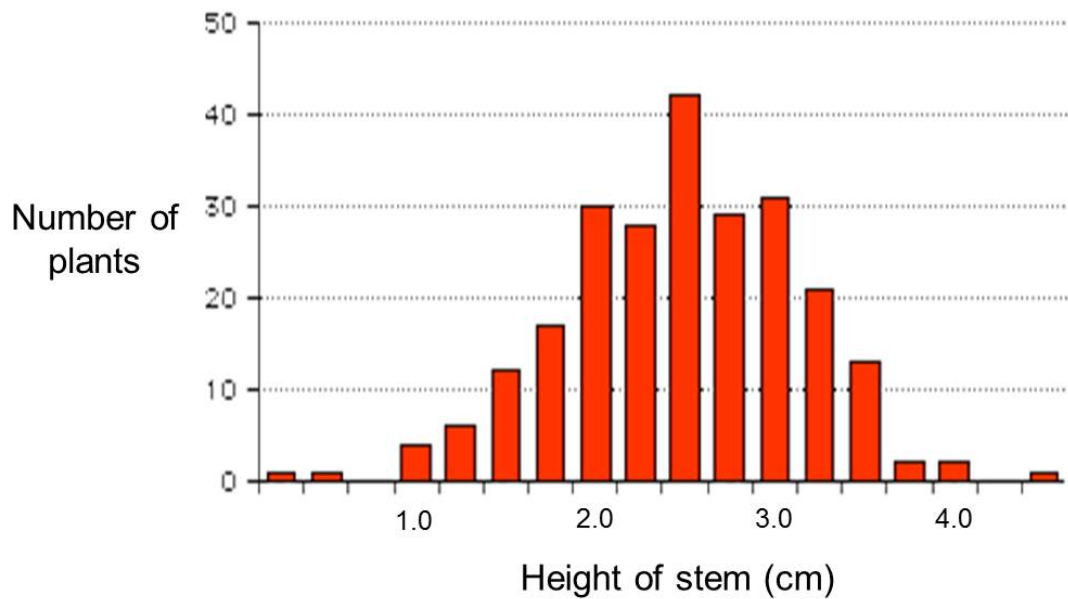


Fig. 4.1

- (i) State the type of variation shown in Fig. 4.1.

[1]

- (ii) Explain why there is a range of phenotypes observed for stem height in *Mimosa pudica*.

[2]

[Total: 9 m]

- 5 The Middle East Respiratory Syndrome (MERS) is a fatal viral respiratory illness caused by a strain of coronavirus called MERS-CoV. The MERS-CoV is a single-stranded RNA virus. Through its spike proteins, this virus infects cells in the respiratory tract in humans, leading to symptoms of fever, cough and shortness of breath, and eventually death.

Fig. 5.1 shows a model of the MERS-CoV structure.

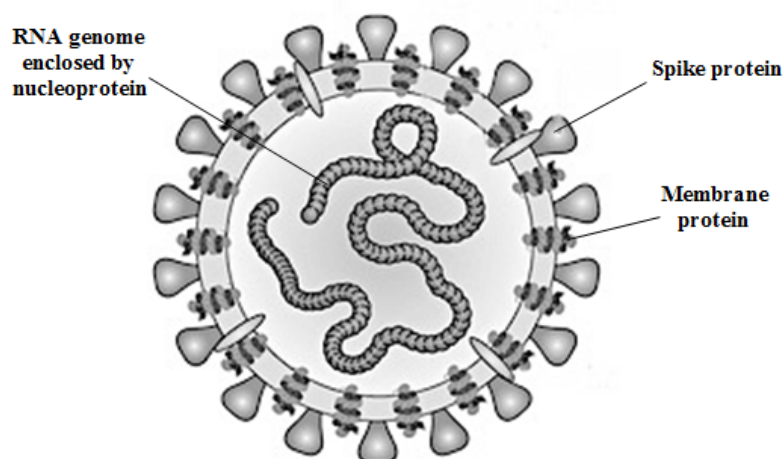


Fig. 5.1

- (a) Explain why coronaviruses mutate faster than DNA viruses.

[2]

Researchers have recently discovered that the entry of MERS-CoV into its host cells is similar to how Human Immunodeficiency Virus (HIV) enters its host cells.

- (b) Suggest how MERS-CoV genome enters its host cell.

[2]

It was previously discovered by the researchers that the cleavage site on the spike protein of MERS-CoV is activated by a host cell protein that subsequently mediates viral entry into host cell. Furin was recently found to be the protein that activates cleavage sites on MERS-CoV.

In a separate experiment, the researchers investigated the cellular furin content and the infection efficiency of MERS-CoV on different cell lines.

Fig. 5.2 shows the relative furin abundance in different cell lines in the host organism.

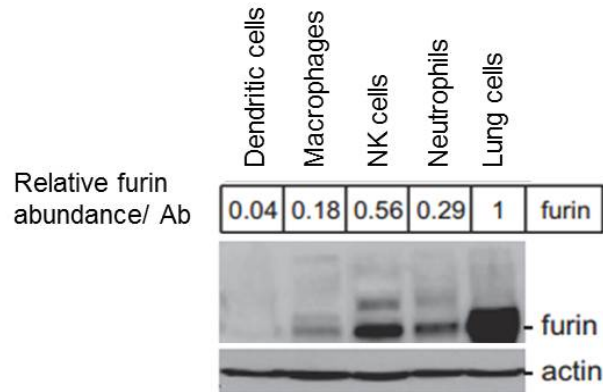


Fig. 5.2

Fig. 5.3 shows the Mers-CoV infection efficiency in the various cell lines investigated.

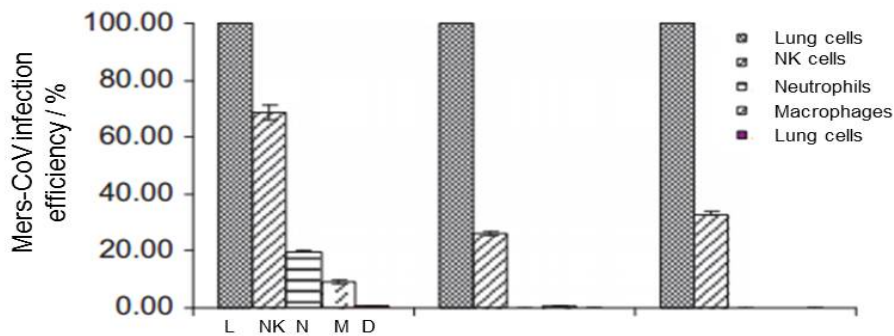


Fig. 5.3

- (c) With reference to Fig. 5.2 and 5.3, explain what can be concluded regarding the cellular furin content and the infection efficiency of MERS-CoV on different cell lines.

[3]

Researchers predict that MERS-CoV is infectious because as it mutates, it is able to bind to different host cell proteins for activation of cleavage site, leading to viral entry.

- (d) Explain the phenomenon that allows MERS-CoV to bind to different proteins for viral entry.

[2]

- (e) Explain why MERS-CoV is considered to be not only a parasite but also an obligatory one.

[2]

[Total: 10 m]

- 6 Fig. 6.1 shows the sequence of events in the insulin signaling pathway in a muscle cell.

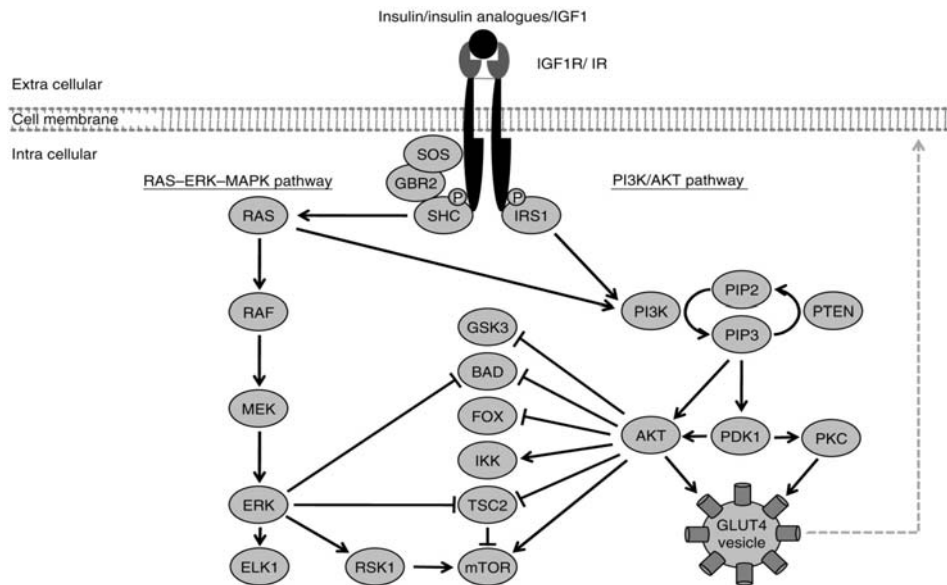


Fig. 6.1

- (a) With reference to Fig. 6.1, explain how a single binding event can trigger two different pathways.

[2]

- (b) With reference to Fig. 6.1, explain how activation of IRS-1 can lead to an increase in glucose permeability of muscle cells.

[3]

- (c) Describe one other mechanism by which IRS-1 can lead to regulation of blood glucose levels.

[2]

When insulin levels in the environment remain high for several hours, cells usually undergo desensitization, such that they no longer respond to that concentration of hormone. This prevents excessive and prolonged receptor activity.

- (d)** Suggest how these cells can decrease the stimulation of receptor molecules by the presence of high concentration of insulin.

[2]

- (e) Describe two advantages of a cell signaling system**

[2]

[Total: 11 m]

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Section C

Answer **all** the questions in this section.

- 7 (a) Acetylcholine is a neurotransmitter that binds to the postsynaptic membrane and stimulates the production of nerve impulses. GABA is another neurotransmitter that is produced by certain neurones in the brain and spinal cord. It binds to the postsynaptic membrane and inhibits the production of nerve impulses.

Fig. 7.1 shows a synapse involving three neurones.

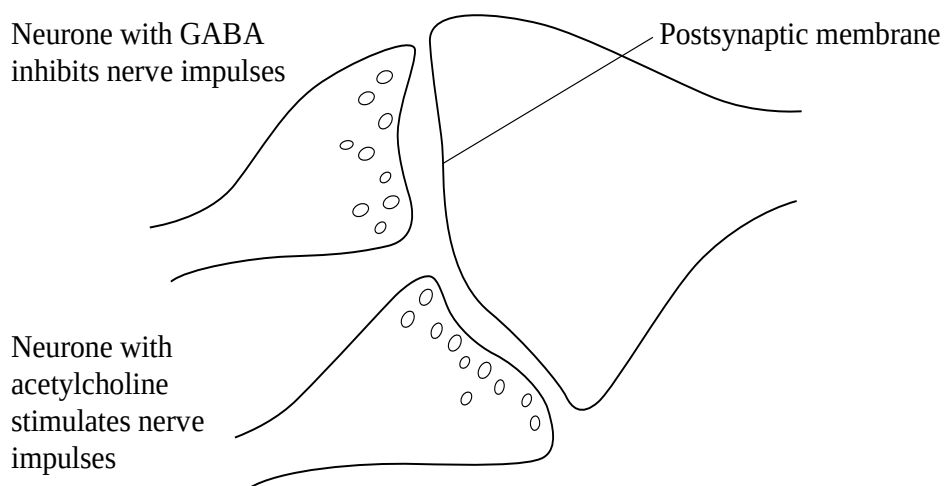


Fig. 7.1

- (i) Describe the sequence of events leading to the release of acetylcholine.

[3]

-
- (ii) State and describe the function of one type of protein molecule on the post-synaptic membrane.

[3]

- (iii) The binding of GABA to receptors on post-synaptic membrane causes negatively charged chloride ions to enter post-synaptic neurone. Explain how this will inhibit production of nerve impulses by post-synaptic neurone.

[2]

- (iv) Epilepsy may result when there is increased neuronal activity in the brain. One form of epilepsy has been linked to an abnormality in GABA receptors. Suggest how an abnormality in GABA receptors may result in epilepsy.

[1]

[Total: 9 m]

- 8 Mitochondrial DNA analysis of various rodents was done and their phylogenetic relationships were shown in Fig. 8.1.

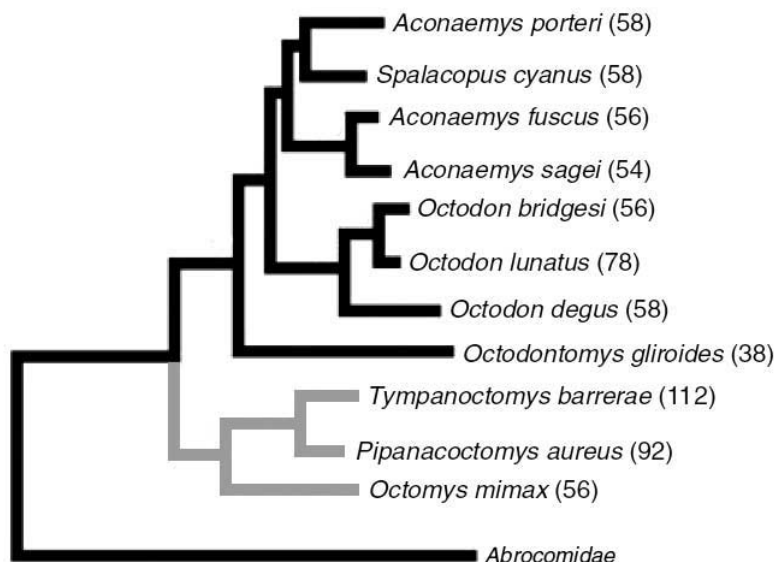


Fig. 8.1

- (a) With reference to Fig. 8.1, state which species is most closely related to *Octodontomys gliroides*.

_____ [1]

- (b) Suggest why mitochondrial DNA was used to study the phylogenetic relationships between the various species of rodents.

 _____ [2]

- (c) Explain how DNA analysis of mitochondrial DNA can provide evidence for the relationships shown in Fig. 8.1.

 _____ [1]

- (d) Explain the difference between classification and phylogeny.

[2]

- (e) Explain two advantages of using molecular evidence compared to anatomical homology in establishing phylogeny.

[2]

- (f) Cases of mammalian tetraploidy are rare. The Plains Viscacha Rat, *Tympanoctomys barrerae*, is one of the few polyploid mammals in existence and it is tetraploid ($4n$) with 112 chromosomes. It is only found in the desert scrublands of the central western plains in Argentina. It was hypothesised that *Tympanoctomys barrerae* evolved from the same common ancestor as *Octomys mimax*, a diploid rodent with 56 chromosomes ($2n$). *O. mimax* inhabits rocky desert and semi-desert environments up to 800 metres above sea level.

Explain how tetraploidy may have first developed in the ancestral rat population.

[2]

[Total: 10 m]

Section D

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

9 (a) Explain how the structure of phospholipids facilitates the process of chemiosmosis in the chloroplast. [10]

(b) Compare and contrast aerobic and anaerobic respiration. [10]

[Total: 20]

10 (a) Define the terms gene and allele and explain how they differ. [4]

(b) Describe the chemical structure of a gene. [8]

(c) Distinguish between mitosis and meiosis. [8]

[Total: 20]

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